

Sudip Parajuli

Software Engineer

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PROFILE

Product-minded software engineer building open-source AI products, dependable backend systems, and full-stack applications. Experience spans model-agnostic AI workflows, document processing, travel platforms, recommendation systems, payments, cloud infrastructure, data engineering, and autonomous robotics. Published with IEEE and active in technical leadership and open source.

EXPERIENCE

2025 - present

Product Engineering

Presenton AI

Builds an open-source AI presentation platform across model orchestration, document ingestion, rich editing, high-fidelity PPTX/PDF export, desktop packaging, and self-hosted deployment. More than 700 contributions to the public repository.

2024 - 2025

Full-stack Engineer

Exploreden Australia - remote

Developed a travel-booking platform using Django and React. Integrated hotel and flight providers, payments, authentication, responsive product flows, and production releases while helping coordinate technical delivery.

2023 - 2024

Backend Engineer

Kinu Tech

Progressed from junior to backend engineer. Built Django services and travel APIs, recommendation and payment integrations, and performance improvements using caching, Celery, AWS Lambda, CDK, ECR, and GitHub Actions.

2024 - 2025

University Lead, CS50 AI

CS50x Nepal

Led AI cohorts, mentored more than 60 learners, and helped organize workshops and X-Hack 3.0, a national AI hackathon with more than 100 participants.

2022 - 2025

Advisor; Secretary; Vice Secretary

EXCESS, IOE Purwanchal Campus

Led programs across Django, IoT, robotics, ROS 2, and web development; organized a 17-team hardware hackathon; and helped deliver technical sessions reaching more than 1,500 students.

2021 - 2025

Data Fellow

Code for Nepal

Built foundations in Python, SQL, data analysis, visualization, machine learning, warehousing, cloud computing, Docker, and collaborative software development.

EDUCATION

2021 - 2025

B.E. Electronics, Communication and Information Engineering

Tribhuvan University - Institute of Engineering, Purwanchal Campus

Coursework and projects across software engineering, data, electronics, communications, embedded systems, machine learning, and robotics.

PUBLICATION

Navigation of Mobile Robot with Nav2 and SLAM Using LiDAR

IEEE - 4th International Conference on Sentiment Analysis and Deep Learning, 2025

Co-authored a peer-reviewed study based on a custom differential-drive robot using ROS 2, Nav2, SLAM Toolbox, LiDAR, Arduino, and behavior trees. DOI: 10.1109/ICSADL65848.2025.10933029.

SELECTED PROJECTS

Presenton

TypeScript - Next.js - Python - LLMs - Docker

Open-source AI presentation generator and API supporting document ingestion, custom templates, multiple models, editable exports, and self-hosting.

docx2everything

Python - PyPI - document processing

Dependency-free package for extracting Markdown, text, tables, images, footnotes, comments, and formatting from DOCX files.

nexREC

PyTorch Geometric - Milvus - Django - PostgreSQL - Remix

Full-stack movie recommendation system built around graph neural networks and vector search.

Clipr

Django REST Framework - Next.js - JWT - Docker

URL-shortening product with authentication, custom links, click tracking, analytics, and a complete dashboard.

dsh-chatgpt-codex

TypeScript - OAuth 2.0 - PKCE - AI agents

ChatGPT OAuth and Codex-model integration for DeepSeek Harness through browser callback and device-code flows.

QuoteReels

Python - Flask - LLMs - KittenTTS - MoviePy

AI-assisted short-form video studio with script generation, stock-footage search, text-to-speech, and a drag-and-drop editor.

Height Comparison

Django - machine learning - web scraping

Backend, prediction model, and data pipeline built from roughly 50,000 celebrity and fictional-character records.

Automatic Emergency Response System

ESP32 - C/C++ - MQTT - Node-RED

IoT prototype for emergency detection, telemetry, and alert notification.

TECHNICAL SKILLS

Languages

Python - TypeScript - JavaScript - SQL - C++ - C - Shell

Backend & APIs

Django - Django REST Framework - FastAPI - Flask - Node.js - Celery - OAuth 2.0 - REST APIs

Frontend

React - Next.js - Remix - Nuxt - Tailwind CSS - Shadcn UI - HTML - CSS

AI & data

PyTorch - PyTorch Geometric - Hugging Face - pandas - NumPy - scikit-learn - GNNs - vector search - LLM tooling

Infrastructure

PostgreSQL - Redis - Milvus - MongoDB - Docker - Nginx - GitHub Actions - AWS CDK/Lambda/EC2/ECR/Lightsail

Robotics & automation

ROS 2 - Nav2 - SLAM - LiDAR - Arduino - ESP32 - MQTT - Selenium - Scrapy - BeautifulSoup

ADDITIONAL WORK

Other projects include DesignClaw, All-in-One Video Downloader, ByaparLink Dashboard, NayaRojgar, Expenses Manager, ICRTAI 2025, CS50x Nepal Smart Attendance, CS50 AI Class Visualizations, Lenovo Energy Management for Linux, Medium Unlocker, and a genetic-algorithm cryptarithmic solver. Full archive: parajulisudip.com/np/projects/